A deterministic emulator of the Veros ACC last-year state

28 October 2026

Summary

- Goal: predict the last-year temperature of the Veros ACC channel directly from the EKE-closure coefficients (c_k, c_ϵ), skipping the 100-year integration.
- Method: a small deterministic network maps $z = (\log c_k, \log c_\epsilon)$, standardised, to the leading PCA coefficients of the training states; it is an ensemble of 5 initialisations. Hyper-parameters come from repeated cross-validation with the one-standard-error rule. A fit takes seconds on a CPU.
- Interpolation (scattered points, a band of three rows, the 5×5 and 7×7 centre blocks, leave-one-out): RMSE 0.017–0.038 K.
  - The nearest training run gives 0.038–0.118 K.
  - The best classical emulator, a Gaussian process on one collapsed parameter s, gives 0.019–0.045 K; the network beats it on all five evaluations (paired test, $p \le 0.04$).
- Extrapolation (top row, outer 51, outer 75): 0.049 / 0.067 / 0.114 K.
  - The GP on s gives 0.079 / 0.074 / 0.125 K; the network beats it on outer 51 and outer 75 ($p \le 0.003$), while on the top row most of the gain is one run.
  - The nearest run gives 0.176 / 0.134 / 0.159 K.
  - The warm corner sets the error: 0.16 K one row from the training set, 1.06 K three cells away.
- Robustness: four initialisation sets change these numbers by at most 0.002 K. The choice of regularisation matters more: among the configurations that cross-validation cannot tell apart, the test error spans up to 0.035 K.
- With four training runs the hyper-parameters cannot be chosen, and depending on them the network gets 0.10–0.19 K against 0.120 K for the GP on s (Section 5).

1 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$ (ratio $2^{2/3}$); 100 years, 30-day output; interior grid $30 \times 42 \times 15$ (x, y, z) with a 930-cell ridge.
- Target: $y_r \in \mathbb{R}^{17970}$, the mean temperature of the last 12 outputs (360 days) of run r on the water cells. $S = 35$ psu everywhere, so it is dropped.
- The runs still drift, so the averaging window matters. Over the last 20 years, the RMS change from the first to the last year is 0.16 K, and the last-year mean differs from the 20-year mean by 0.08 K.
- The two parameters act through one combination, $s = \log c_k - 1.08 \log c_\epsilon$ (Figure 1). Below $s \approx -2$ (48 runs) the domain-mean T stays within 0.09 K; above, it rises by 1.1 K.

2 Method

- Input z: $(\log c_k, \log c_\epsilon)$ standardised over the design.
- Output: a field in the span of the training states,

$$\hat{y}(z) = \mu + \sum_{k=1}^{K} a_k(z)\, v_k, \qquad K = \min(20,\, n-1), \tag{1}$$

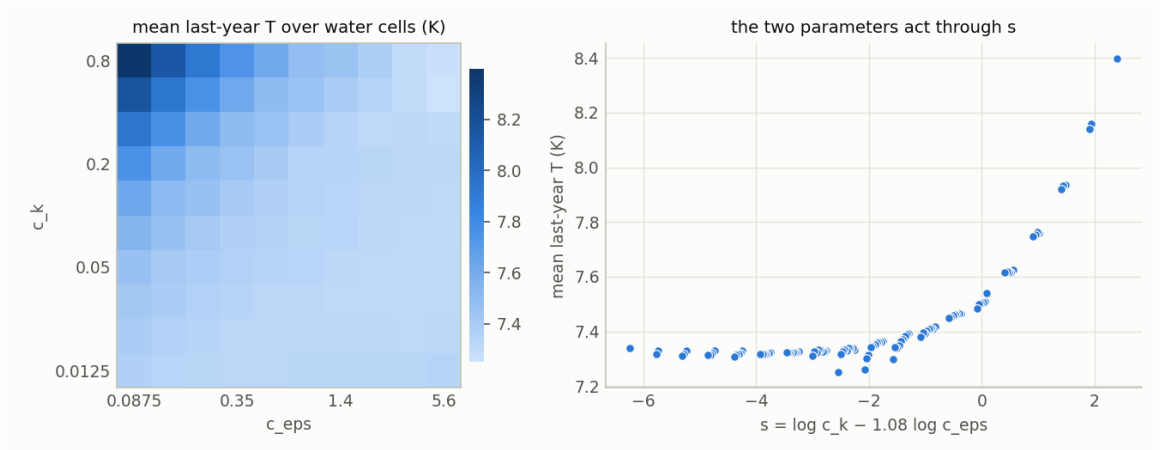


Figure 1: Mean last-year T over the water cells: over the parameter grid (left) and against s (right), the direction along which the first PCA coefficient of the states varies linearly.

where μ and v_k are the mean and the leading principal directions of the n training states. So $K = 20$ on every split except the four-run one ($K = 3$).

- Network $a(z)$: a multilayer perceptron $2 \to 64 \to 64 \to K$ (scikit-learn, L-BFGS). Its targets are $a_k(z_r) = v_k^\top (y_r - \mu)$ divided by $\mathrm{std}(a_1)$, so with orthonormal v_k the data term is the physical MSE of the projected field up to a constant. In scikit-learn's normalisation:

$$\mathcal{L} = \frac{1}{2nK} \sum_{r=1}^{n} \frac{\|a(z_r) - V^\top (y_r - \mu)\|^2}{\mathrm{std}(a_1)^2} + \frac{\alpha}{2n} \|W\|^2. \tag{2}$$

- Ensemble: the prediction is the mean of 5 networks that differ only in their initialisation. The whole evaluation is repeated with 4 such sets; reported values average them, with their range in brackets.
- Hyper-parameters: $(\text{activation}, \alpha) \in \{\text{ReLU}, \tanh\} \times \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, chosen on the training runs of each split.
  - The search is 5-fold cross-validation repeated 3 times.
  - The one-standard-error rule then picks the most regularised configuration whose CV RMSE is within one standard error of the best.
  - The choice is ReLU with $\alpha = 10^{-2}$ on every split except outer 75 ($\alpha = 10^{-1}$) and four runs ($\alpha = 1$).
  - This rule replaced a single 5-fold cross-validation that took the minimum. An independent review showed that choice was decided by fold noise. The new rule is applied unchanged to every split.
  - For leave-one-out the choice is made once, on all 100 runs. Each held-out run therefore takes part in it: a mild selection leak, with flat scores.
- Fitting on run means rather than on snapshots changes nothing: $\sum_t \|f(z_r) - x_{r,t}\|^2 = n_t \|f(z_r) - \bar{x}_r\|^2 + \text{const}$.
- Baselines, none tuned by cross-validation (unlike the network):
  - the training mean;
  - the nearest training run in z;
  - thin-plate RBF interpolation of every cell (exact, no smoothing);
  - a GP per PCA coefficient on z (Matérn 5/2, one length scale per input, white noise up to the full variance; hyper-parameters by maximum marginal likelihood);
  - the same GP on $s = w^\top z$, with w from a linear fit of a_1 on z.

3 Evaluation

- Eight hold-out splits of the grid, plus leave-one-out over the 100 runs:
  - interpolation: 10 scattered interior runs; the rows $c_k = 0.126$, 0.2, 0.3175; the centre 5×5 and 7×7 blocks;

  - extrapolation: the top row ($c_k = 0.8$); the 51 or 75 outer runs, training on the centre;
  - four training runs, one step in from each corner (Section 5).
- Metric: RMSE over the water cells against the last-year mean, per held-out run, then the mean and median over the runs.
- Paired comparison with the GP on s, on the per-run difference (network RMSE averaged over the 4 initialisation sets, minus the GP's): its mean, a 95 % bootstrap interval, and a Wilcoxon signed-rank p-value.
- Sensitivity: the test RMSE of every configuration on every split (first initialisation set). It is not used for the selection.
- Cost: a fit takes seconds on one core, a prediction microseconds. The whole evaluation takes about one hour on 8 cores, mostly the repeated cross-validation and the 100 leave-one-out refits.

split	held out / train	network	median	GP on s	GP on z	RBF	nearest	mean state
scattered	10 / 90	**0.017** (0.017–0.017)	0.013	0.019	0.022	0.021	0.038	0.161
band (three rows)	30 / 70	**0.038** (0.037–0.039)	0.033	0.045	0.051	0.057	0.118	0.204
centre 5×5	25 / 75	**0.025** (0.024–0.025)	0.020	0.029	0.035	0.037	0.080	0.162
centre 7×7	49 / 51	**0.029** (0.028–0.030)	0.026	0.035	0.069	0.068	0.116	0.203
leave-one-out	100 / 99	**0.024** (0.024–0.024)	0.017	0.028	0.032	0.034	0.085	0.221
top row	10 / 90	**0.049** (0.047–0.050)	0.033	0.079	0.083	0.084	0.176	0.482
outer 51	51 / 49	**0.067** (0.066–0.067)	0.027	0.074	0.081	0.093	0.134	0.265
outer 75	75 / 25	**0.114** (0.113–0.114)	0.041	0.125	0.131	0.138	0.159	0.226

Table 1: RMSE (K) of the last-year temperature over the water cells, mean over the held-out runs. Network: mean over 4 initialisation sets, with their range in brackets, and the median over runs. Bold: lowest mean per split. The four-run split is in Section 5.

	network minus GP on s			configurations within one standard error	
split	difference [95 % interval]	Wilcoxon p	wins	test RMSE (number)	below GP on s
scattered	−0.0026 [−0.0054, −0.0006]	0.037	7/10	0.016–0.017 (5)	5
band (three rows)	−0.0074 [−0.0132, −0.0019]	0.015	23/30	0.036–0.041 (6)	6
centre 5×5	−0.0044 [−0.0071, −0.0018]	0.004	17/25	0.025–0.029 (5)	5
centre 7×7	−0.0059 [−0.0096, −0.0024]	0.002	35/49	0.029–0.045 (6)	3
leave-one-out	−0.0046 [−0.0086, −0.0015]	< 0.001	67/100	–	–
top row	−0.0300 [−0.0769, −0.0006]	0.19	7/10	0.050–0.060 (5)	5
outer 51	−0.0075 [−0.0123, −0.0031]	0.003	36/51	0.067–0.076 (5)	3
outer 75	−0.0111 [−0.0170, −0.0059]	0.001	51/75	0.113–0.148 (6)	3

Table 2: Left: paired comparison of the network with the GP on s, per held-out run (K; negative = network better). Right: test RMSE of the network over the configurations whose CV RMSE is within one standard error of the best (first initialisation set), and how many of them beat the GP on s.

4 Findings

- **In interpolation the network beats every baseline** (Tables 1 and 2, Figure 2).
  - Its errors are a third of the nearest run's.
  - Against the GP on s the margin is 0.003–0.007 K, significant on all five evaluations (p from 0.037 to < 0.001).
- **In extrapolation it is better on outer 51 and outer 75** ($p = 0.003$ and 0.001). On the top row the gain (0.049 vs 0.079 K) is mostly the warm-corner run (0.16 vs 0.37 K); without it the network gets 0.037 vs 0.046 K and wins 6 of 9, and the test is not significant ($p = 0.19$).
- **The result depends on regularisation, not on initialisation** (Table 2, right).
  - Four initialisation sets change a split's mean by at most 0.002 K.
  - Among the configurations cross-validation cannot tell apart, the test error spans up to 0.035 K. On the 7×7 block, outer 51 and outer 75, only half of them beat the GP on s.

  - On those three splits and the 5×5 block, the most regularised configuration (the rule's choice) is the best of the set; on the top row it is nearly the best; on scattered and band it is mid-range or above.
- **The error sits in the upper** 500 m (Figures 3 and 4). In interpolation it is 0.014–0.055 K there and 0.003–0.016 K below 1000 m. At the centre of the 7×7 block, the predicted anomaly (relative to the training mean) has correlation 0.98 with the true one and amplitude 1.07.
- **The warm corner** ($c_k = 0.8$, $c_\epsilon = 0.0875$) is the hardest state (Figure 4).
  - One row from the training set (top row), the network reproduces its warming: correlation 0.99, amplitude 0.99, 0.16 K.
  - Three cells away (outer 75), it keeps a weak version of the pattern (correlation 0.62) with a third of the amplitude: a cold bias of −0.68 K and 1.06 K RMSE.
  - Every method is between 1.06 and 1.45 K there.
- **Part of the extrapolation error is the basis.** Projecting the held-out truth on the training principal directions already leaves 0.010–0.017 K in interpolation and 0.031–0.079 K in extrapolation. The outer-75 value (0.079 K of the network's 0.114 K) is mostly the corner run, where the floor is 0.85 K.
- **The spread over initialisations is not an uncertainty.** It is 0.003–0.009 K in interpolation and 0.008–0.010 K in extrapolation, while the error grows from 0.017 to 0.114 K.
- The comparison is between a 5-member ensemble and single baselines, and only the network's hyper-parameters are tuned by cross-validation.

5 A failure case: four training runs

- Training on $c_k \in \{0.0198, 0.504\} \times c_\epsilon \in \{0.139, 3.53\}$ leaves 96 runs to predict, with $K = 3$ principal directions (2 inside the cross-validation folds).
- Cross-validation cannot choose here. Its standard error (0.15 K) covers six configurations, and the rule picks $\alpha = 1$.
- With that choice the network gets 0.147 K (0.134–0.169 over the initialisation sets). The GP on s gets 0.120 K, the RBF 0.146 K, the nearest run 0.181 K.
- Over the ten configurations the network ranges from 0.10 to 0.19 K: the outcome is set by a choice that four runs cannot inform.
- Regimes:
  - where the state saturates ($s < -2$, 46 runs), the medians are 0.040 (nearest run), 0.058 (network) and 0.061 K (GP on s);
  - on the warm side (50 runs) they are 0.197 (network), 0.172 (GP on s) and 0.223 K (nearest run).
- With $\alpha = 1$ the network is too flat (Figure 4, rows 2 and 5).
  - At the centre run its anomaly has the right pattern (correlation 0.98) with 45 % of the amplitude: 0.12 K, against 0.036 K for the GP on s.
  - At the corner: correlation 0.86, amplitude 0.40, 0.80 K.

6 Next steps

- Inputs $(s, s_\perp)$ instead of $(\log c_k, \log c_\epsilon)$, to build in the collapse that makes the GP on s strong with few runs.
- For extrapolation, either a decoder not restricted to the span of the training states, or more runs near the warm corner.
- An uncertainty from resampling the training runs rather than the initialisations.
- Tune the baselines by cross-validation as well (RBF smoothing, GP noise and mean function), for an even comparison.
- A choice of regularisation that works with very few runs, e.g. fixed in advance from the larger splits.

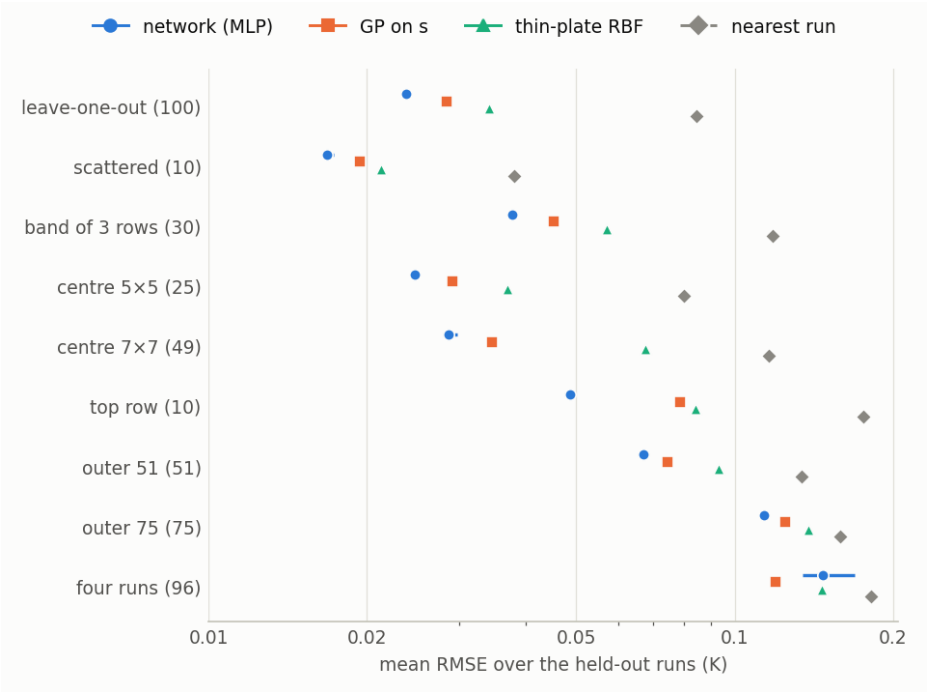


Figure 2: Mean RMSE over the held-out runs per split (log scale) for the network and three baselines: the GP on s, the thin-plate RBF and the nearest run. Held-out run counts are in brackets. Network: mean over 4 initialisation sets; the bar is their range.

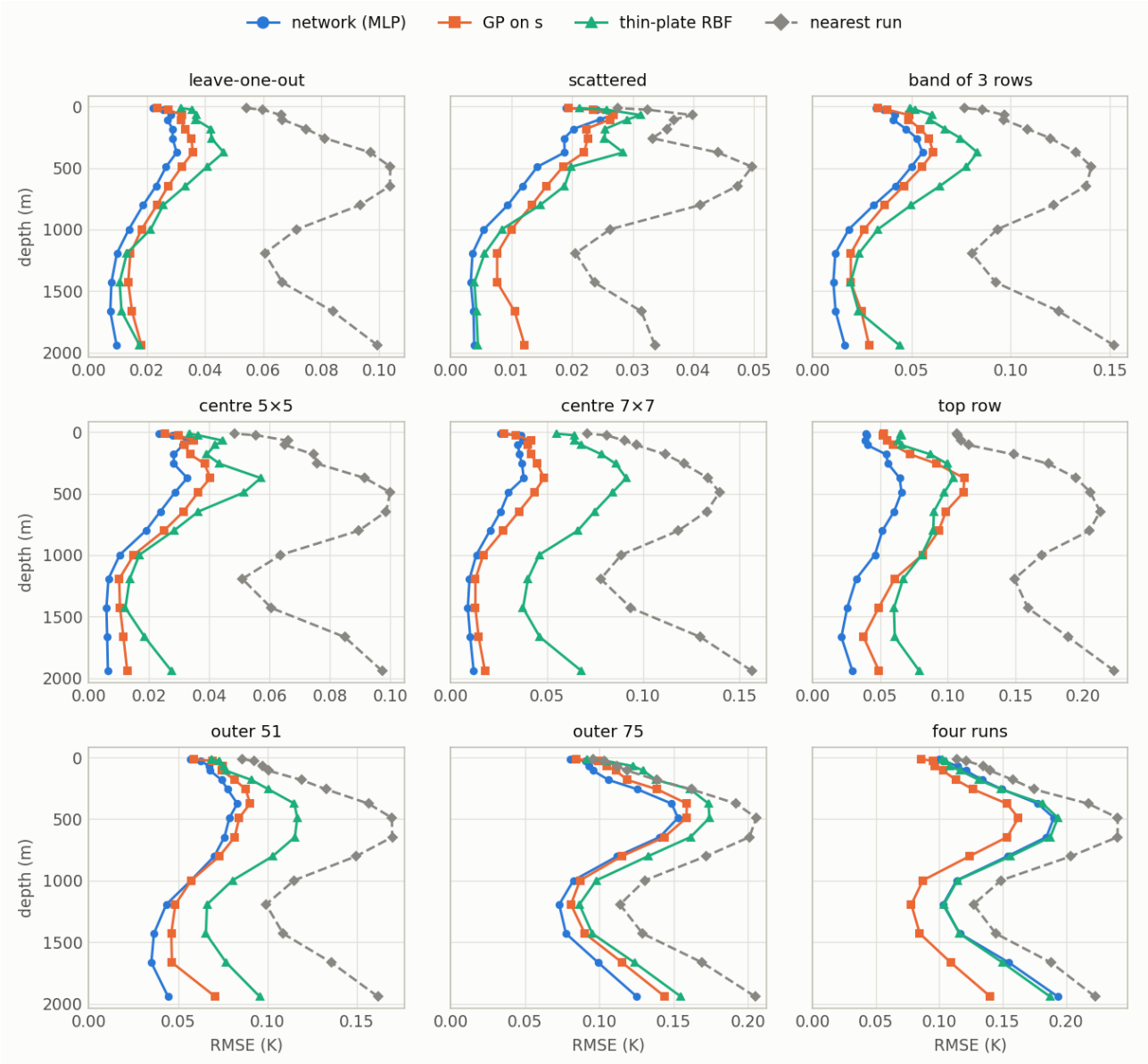


Figure 3: RMSE by depth, mean over the held-out runs of each split (network: averaged over the initialisation sets).

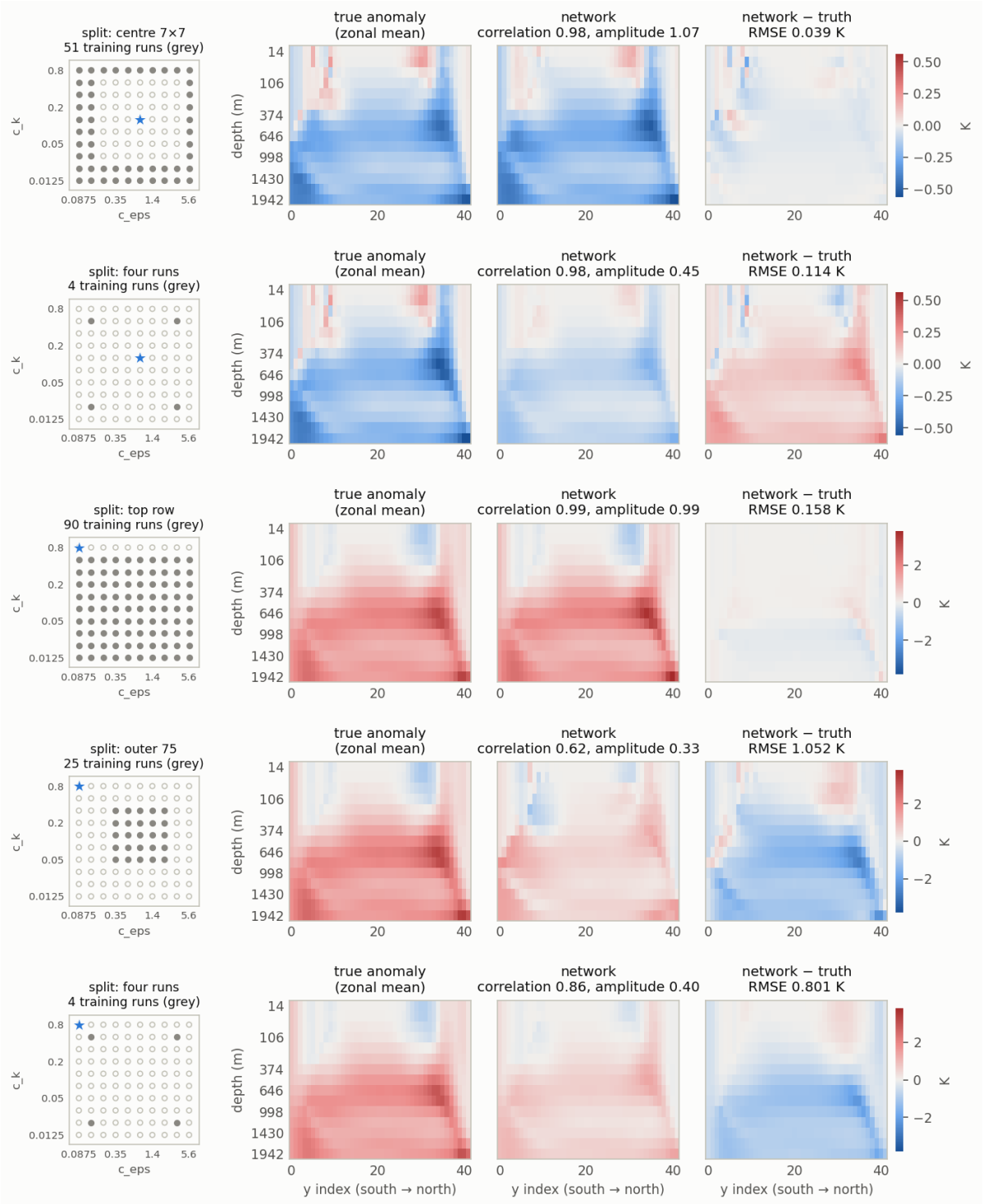


Figure 4: Zonal-mean sections of the last-year anomaly, relative to the mean of the split's training runs (what the network has to add to it): truth, network (first initialisation set) and their difference. Each run has one colour scale, and levels are drawn with equal height. Left: the split (grey: training runs; open circles: held out; star: the run shown). Rows 1–2: the centre run of the 7×7 block, held out with the block (51 training runs) and in the four-run split. Rows 3–5: the warm-corner run, held out with the top row (90 training runs), with the outer 75 (25) and in the four-run split (4). Correlation and amplitude (the regression slope of the predicted on the true anomaly) are over all water cells.